MATH329 · OPERATIONS RESEARCH

# Midterm 471 Practice

60 minutes

**Muharib**

## 01 Questions

Answer each question in the space below it. The full solution is in a separate file.

**Q1** TRUE / FALSE · 5 M

Answer true or false for all of the following:

1. All the parameters (coefficients of the objective function and constraints) of the model are unknown.
2. A basic feasible solution is a feasible solution to LPP which is also the basic solution.
3. A basic feasible solution is called degenerate if value of at least one basic variable is zero.
4. If all the ratios in simplex table are negative or infinite then the solution an alternative.
5. The solution of the LPP may be compatible with its constraints imposed by the available resources.

**Q2** GRAPHICAL · 9 M

Solve the following problem graphically:

Maximum $z = 3x_1 + 2x_2$

S.to: $2x_1 + x_2 \leq 10$

$x_1 + x_2 \leq 8$

$-x_1 \geq -4$

$x_1 \geq 0, x_2 \geq 0$

**Q3** STANDARD FORM + SIMPLEX

Given the following problem:

Maximize $Z = 2x_1 + 3x_2$

S.to $x_1 + 2x_2 \leq 16$

$2x_1 + x_2 \leq 18$

$x_1 \geq 0, x_2 \geq 0$

a) Write the problem in standard form (equation form).

b) Solve the LP problem using the Simplex method.

| Basic | $x_1$ | $x_2$ | $s_1$ | $s_2$ | Sol. |
|---|---|---|---|---|---|
| Z | | | | | |
| | | | | | |
| | | | | | |
| Z | | | | | |
| | | | | | |
| | | | | | |
| Z | | | | | |
| | | | | | |
| | | | | | |

The paper is pasted inside the lecture notes without its header, so it has no date. I call it 471 because the slides around it are marked 471. The marks of the third question are not shown.